

Lecture 19

Monday, April 20, 2026 7:22 PM

Recap: Naive Bayes

- ① Core assumption : Features conditionally independent given class label
- ② Model : $P(x|y) = \prod_{j=1}^n P(x_j|y)$
- ③ Parameters : $\phi_y, \phi_{j|k} = P(x_j=1 | \text{class}=k)$
- ④ MLE : one pass, closed form
- ⑤ Laplace smoothing to avoid zero probabilities

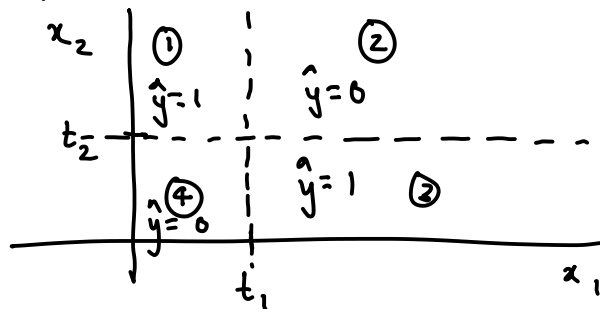
Today: Decision Trees

Limitations:

$$- \theta^T x = 0$$

Example A

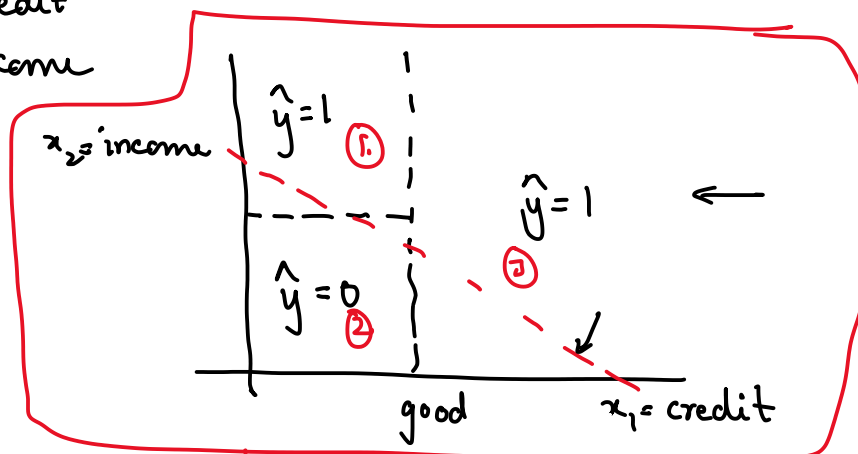
- ① $y=1$ if $(x_1 < t_1 \text{ \& } x_2 > t_2)$ or $(x_1 > t_1 \text{ and } x_2 < t_2)$



feature interaction

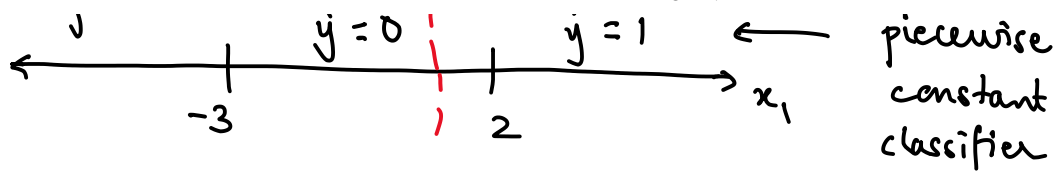
- ② Example B:

→ "Income is relevant only when credit is bad"

 $x_1 = \text{credit}$ $x_2 = \text{income}$ 

③

 $\hat{y}=1$



Other drawbacks of logistic regression, naïve Bayes, GDA

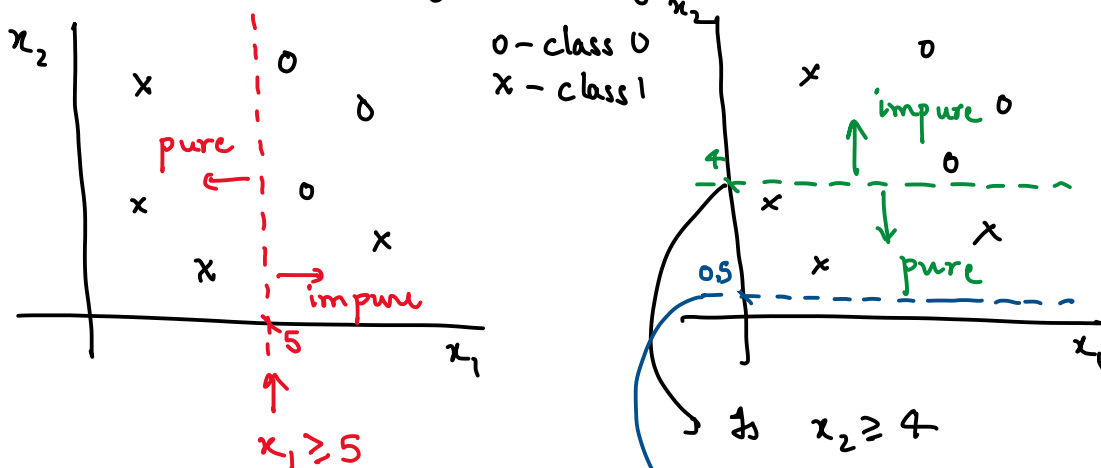
Formal Definition of a Decision Tree

- ① Binary tree
- ② Each internal node store a "split question"
is $x_j \geq s$
 ↑ ↑
 dimension split value
- ③ Each leaf node gives a predicted class.

Note: $\left. \begin{array}{l} \# \text{ of parameters is not fixed} \\ \# \text{ of internal is not fixed.} \end{array} \right\}$ need to be determined during training

Recall: parametrized, Likelihood, argmax likelihood

How to measure the goodness of a split?



Gini Impurity measures the purity or goodness of a partition.

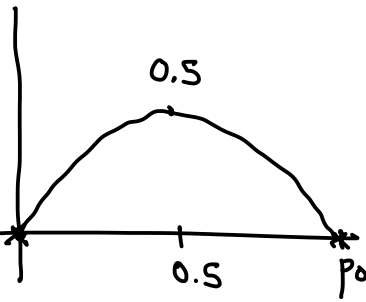
Let S be the set of samples in a group.
 Let $p_k = \frac{|S_k|}{|S|} \rightarrow$ class k samples in the group
 ↓
 total samples in the group

fraction of class k ~~same~~ samples in the group

$$G(s) = 1 - \sum_k p_k^2 \quad \leftarrow$$

Binary classes. $\rightarrow p_0 = 1 - p_1$ $G(s)$
 $G(s) = 2 p_0 (1 - p_0)$

$$p_0 = \begin{cases} 1 & \text{all samples are class 0} \\ 1/2 & \text{50\% samples are class 0} \\ 0 & \text{all samples are class 1} \end{cases}$$



$G = 0$ if all samples have the same class

$G = 0.5$ if 50% are from each of the two classes

What a tree is ✓

How to use it ✓

How to build it?

Training data (recall)

$$\{x^{(i)}, y^{(i)}\}_{i=1}^N \quad x^{(i)} \in \mathbb{R}^d \quad y^{(i)} \in \{0, 1, \dots, K-1\}$$

Plain english explanation : d (minimize set size)

- ① $I = \{1, 2, \dots, N\}$
- ② Is this node pure? - create a leaf, declare majority class as the predictor
 Is this set small enough? $N < d$ - create a leaf, declare majority class as predictor

③ Find the best split :

try every feature, every split value

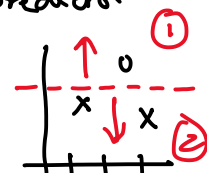
For every feature / split value

Compute the Gini impurity of the two groups.

Compute the weighted gini impurity

Pick feature / value with the lower w. G. I.

(more coming later)



④ ~~Partition~~ Partition the samples: $x_j \geq s$
 $x_j < s$

⑤ Apply ① to ④ on each of the two groups

Recursive process

